WaveCatBoost: Hybrid Air Quality Forecasting

Applied Time Series Analysis — Course Project

Group 8

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Problem and Approach

Why this is hard

- Air-quality forecasting needs **near real-time** answers.
- That pushes us toward **lighter** models (trees, short lags).
- The series is **non-stationary**: spikes, regime shifts, and changing variance.
- Light models struggle on the **raw** mix of spikes, diurnal cycles, and slow drift.

Paper approach

1. Split the series into scale-wise components (details + smooth) with Haar MODWT.
2. Forecast each component with CatBoost.
3. Reconstruct by summing bands; optionally add conformal intervals.

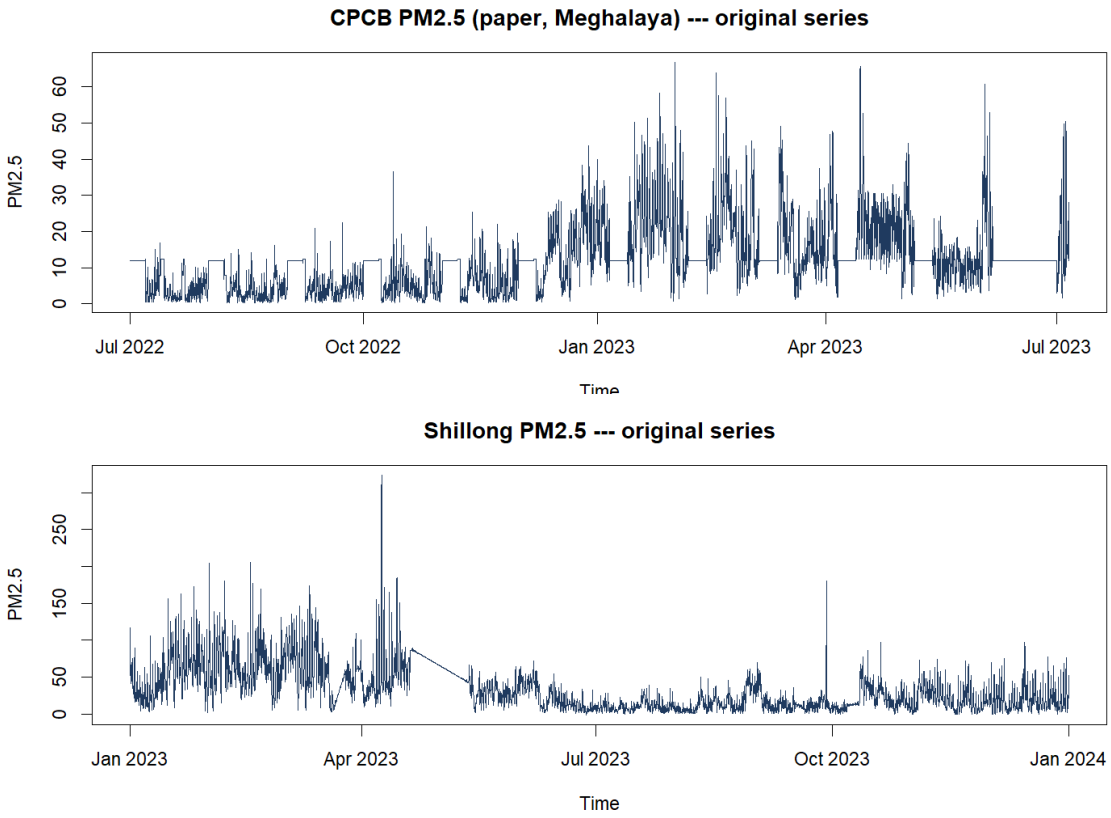
Datasets

Paper CSVs (Meghalaya, hourly)

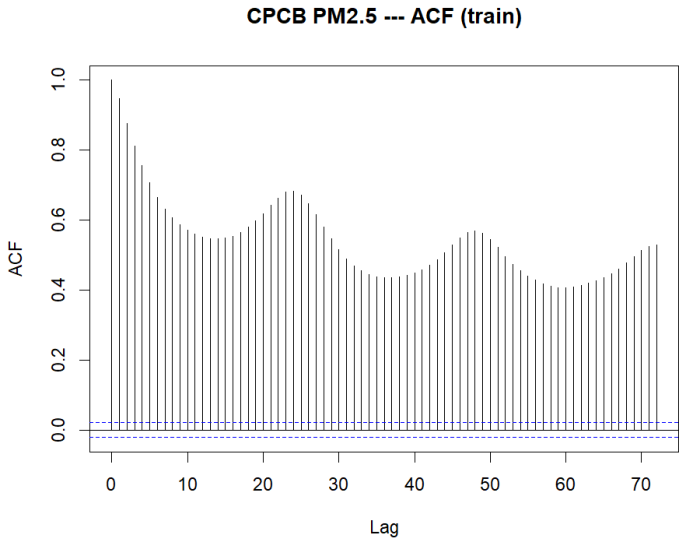
- **CPCB**, **ID1** from Borah et al.
- $n \approx 8880$ / 9505.
- NO_2, O_3, CO, SO_2, $PM_{2.5}$, PM_{10}.

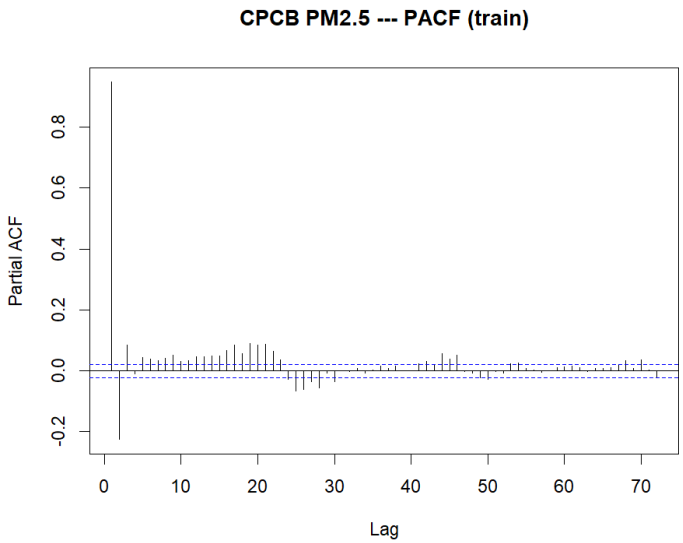
Extra CAAQMS (15-min $\rightarrow$ hourly)

- **Shillong:** Lumpyngngad, JN Stadium.
- Delhi: ITO, R K Puram.



Dataset diagnostics — ACF / PACF





ACF

Correlation of Y_t with Y_{t-k} for lags up to 72.

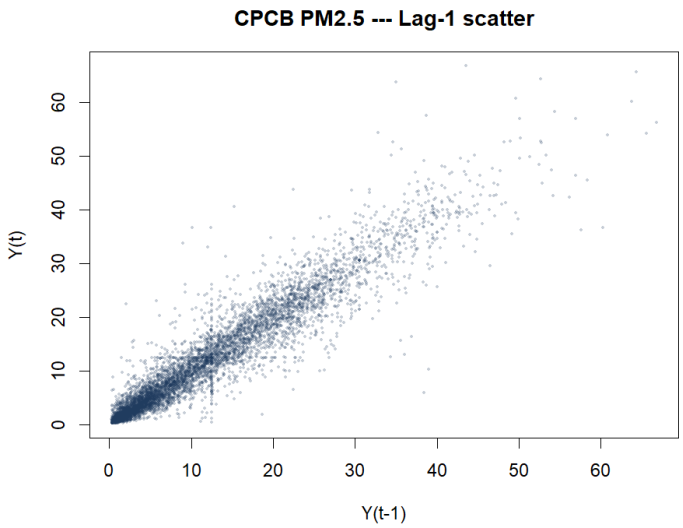
Stays high and waves every ~ 24 lags $\Rightarrow$ strong daily seasonality plus long memory (yesterday's level still matters).

PACF

"Extra" correlation after earlier lags are accounted for.

Huge spike at lag 1, then mostly small $\Rightarrow$ strongest short-term link is the previous hour (AR-style / lag-1 features). Lag 24 shows more in the ACF than as a big PACF spike.

Dataset diagnostics — lag scatters

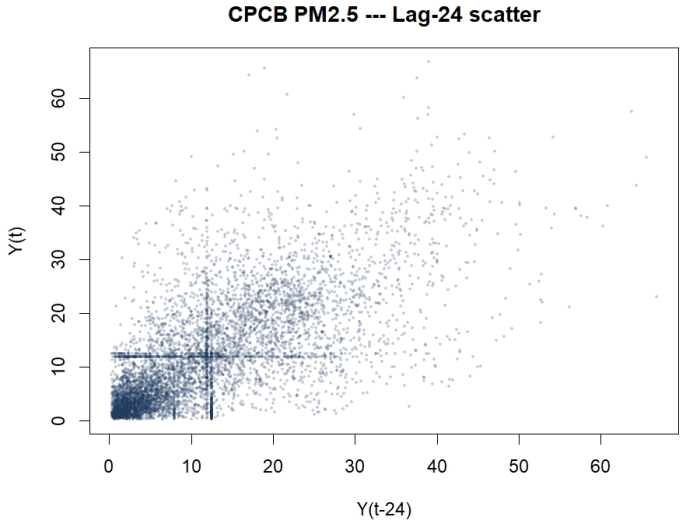


Lag-1 scatter

Y_t vs Y_{t-1}: tight diagonal.

Next hour is usually close to this hour

$\Rightarrow$ high persistence / smooth hour-to-hour changes.

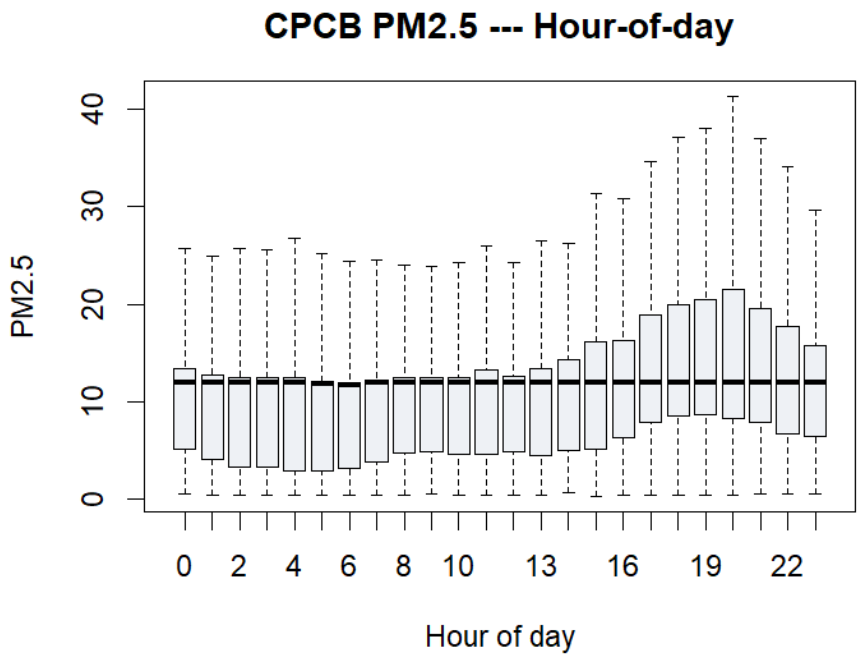


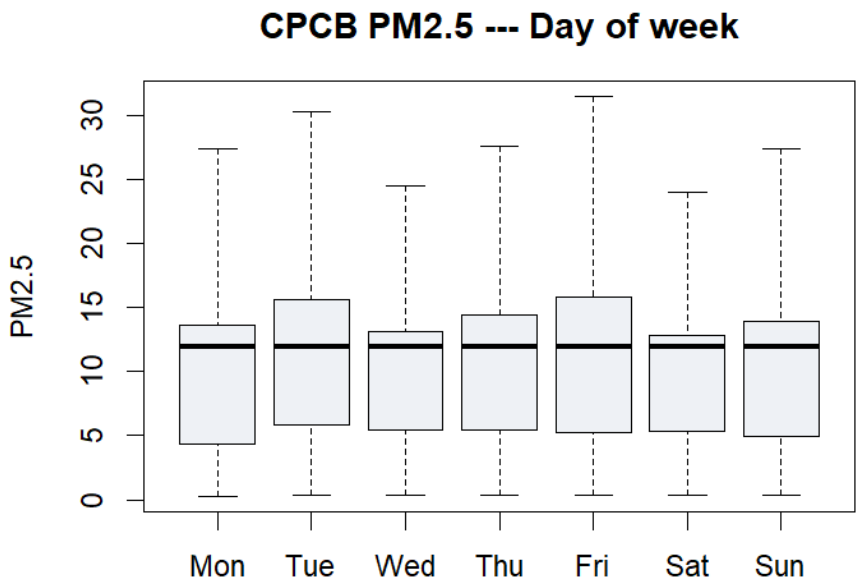
Lag-24 scatter

Y_t vs Y_{t-24}: still positive (same hour yesterday helps), but much looser than lag 1.

Daily cycle is real, but weaker than the immediate lag.

Seasonal structure — CPCB $PM_{2.5}$





- Clear diurnal pattern; milder weekday effect.
- S = **seasonality period** (hours in one cycle). Here $S{=}24$ = one day.
- Used for seasonal naive, SARIMA seasonal lag, and the MASE denominator.
- Also motivates lag features up to 24 h.

What we did before modelling

- **Hourly grid:** end-labelled hourly means (15-min CAAQMS $\rightarrow$ hour t).
- **Quality screen:** physical bounds, stuck sensors, spike flags.
- **Causal fill:** impute only with information available up to the cut time.
- **Chronological split:** train $\rightarrow$ val (14 d tuning) $\rightarrow$ test (last h hours); $h \in \{24, 168, 336, 744\}$. No random split.
- **Scale:** min–max to $[0, 1]$ fit on **train only**.
- Paper Meghalaya CSVs are already hourly $\Rightarrow$ skip re-aggregation / fill. Extra stations (Shillong, Delhi) get the full clean path into an isolated store.

Why decompose? Why not Fourier?

Why split the series

- Raw Y_t mixes noise, diurnal swings, and slow level shifts.
- One light model on the mix misses scale-specific structure.
- Decompose $\rightarrow$ forecast each band $\rightarrow$ add back $=$ richer dynamics without a heavy global model.

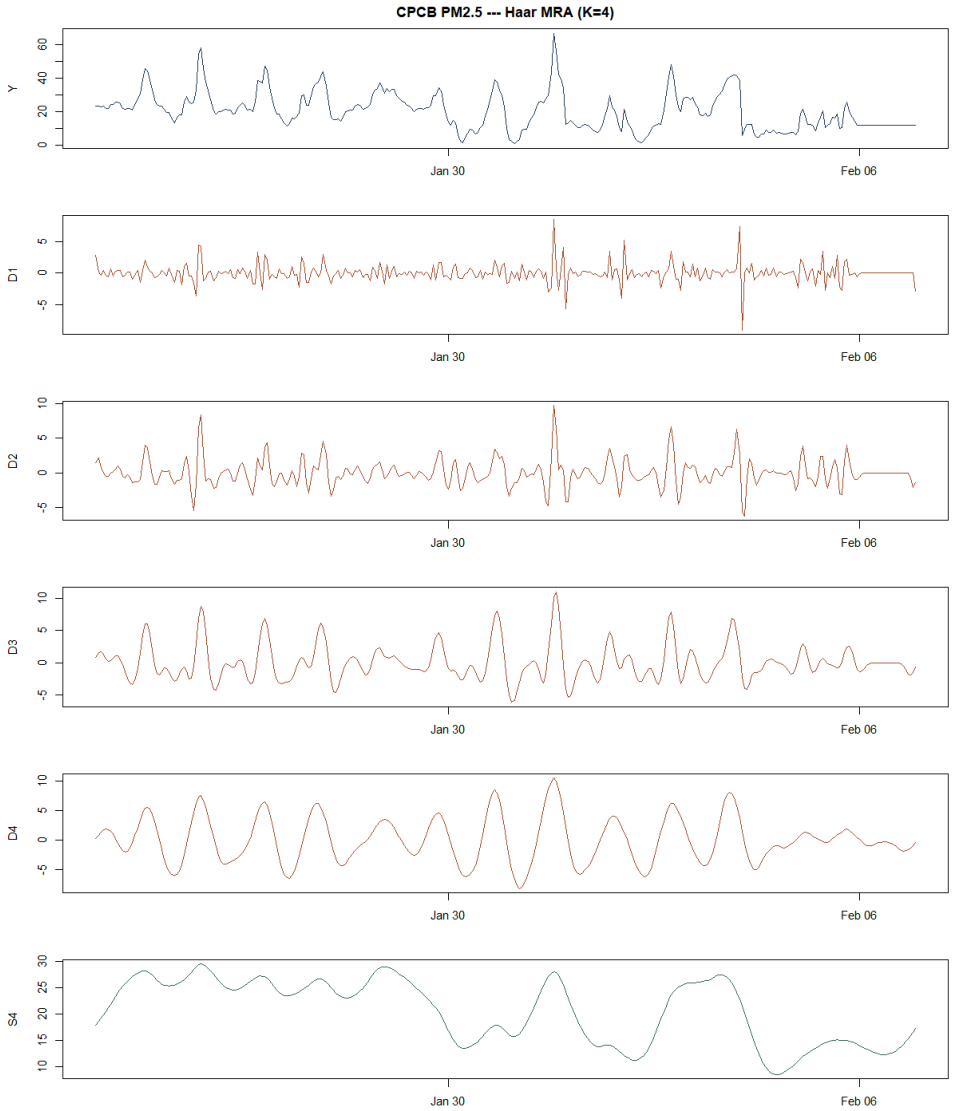
Why not Fourier

- Fourier uses **global** sinusoids over the whole record.
- Air quality is **non-stationary** — spikes and regime changes are local in time.
- Wavelets give a joint **time–scale** view; MODWT keeps every band length N (no downsampling).
- MODWT MRA is additive for Haar, D4, LA8, . . . : $Y_t = \sum_k D_{k,t} + S_{K,t}$ (paper uses Haar).

MODWT: additive components

$$Y_t = \sum_{k=1}^{K} D_{k,t} + S_{K,t}$$

- $D_{k,t}$: detail at scale $\sim 2^k$ h.
- $S_{K,t}$: smooth after K levels.
- Paper: $K \approx \ln N$; we use K=4–5 ($\sim$ 16–32 h smooth).
- D_1: high-frequency component;
  D_k: lower-frequency component;
  S_K: smooth trend / level.



MODWT: mathematical intuition

Wavelet filter: $p_m \to D$ (details)
$\sum_m p_m = 0$ (measures change)

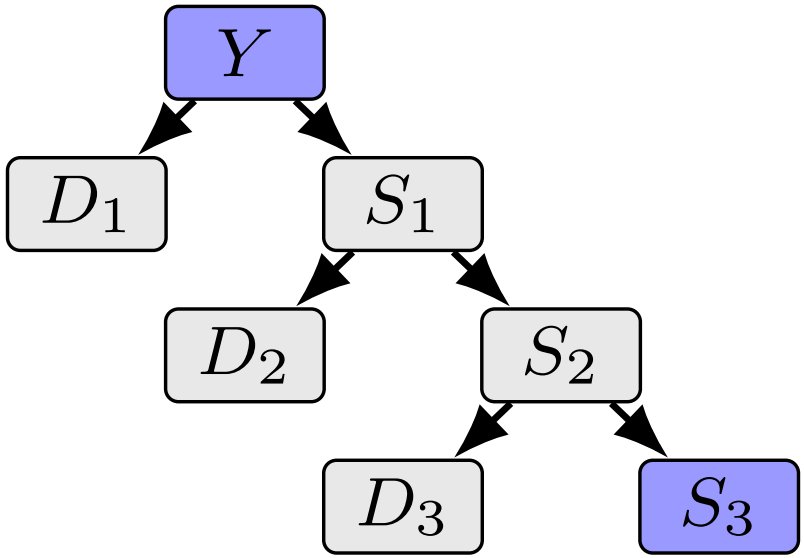
Scaling filter: $q_m \to S$ (smooth)
$\sum_m q_m = 1$ (measures average)

Filter condition (nonzero n):
$\sum_m p_m p_{m+2n} = \sum_m q_m q_{m+2n} = 0$

Recursion at level j ($S_{0,t} = Y_t$)
$D_{j,t} = \sum_m p_m \, S_{j-1,(t-2^{j-1}m) \bmod N}$
$S_{j,t} = \sum_m q_m \, S_{j-1,(t-2^{j-1}m) \bmod N}$

Haar: $q = (\frac{1}{2}, \frac{1}{2})$, $p = (\frac{1}{2}, -\frac{1}{2})$.

Tap spacing doubles each level.

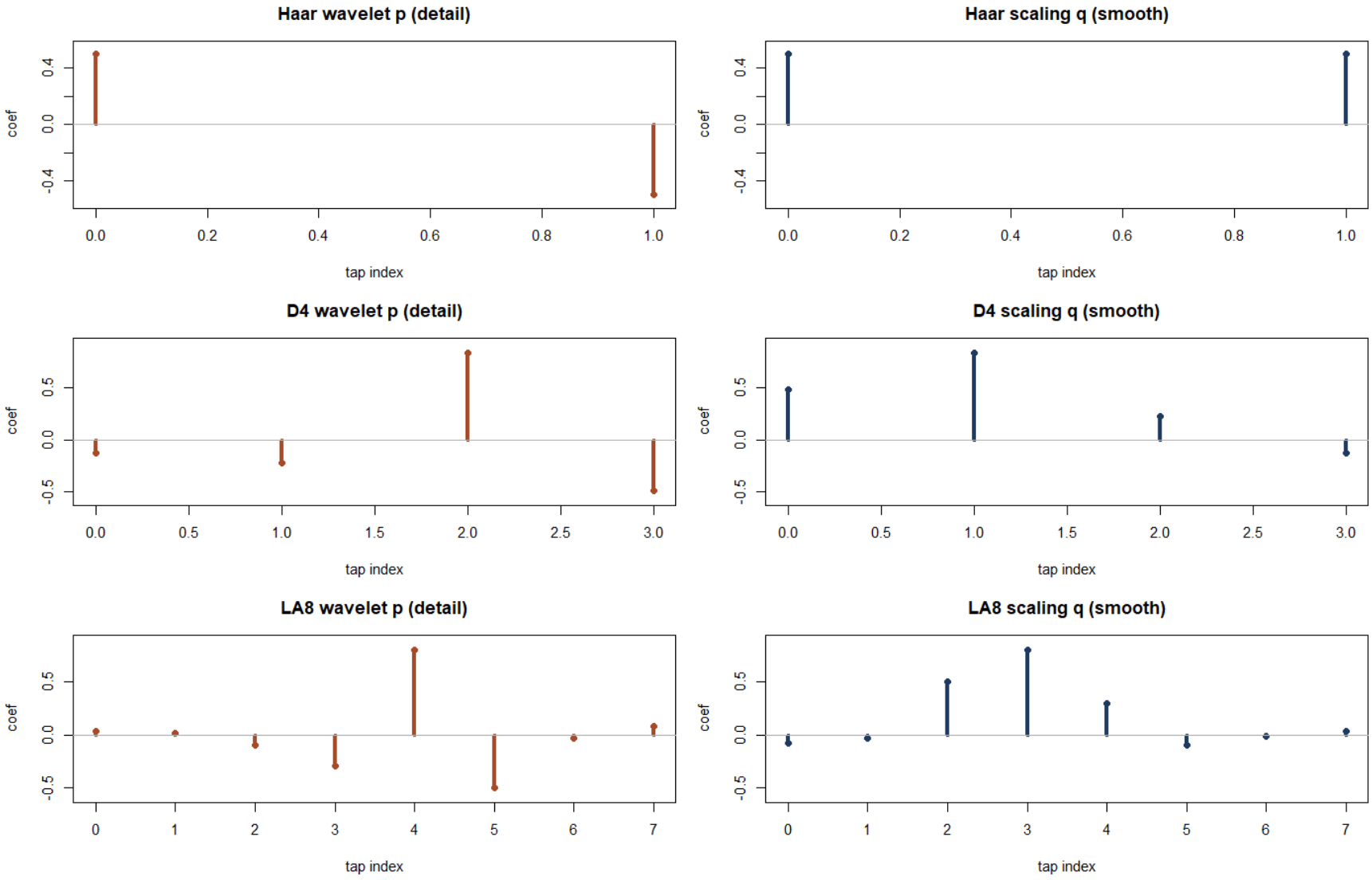


Only the smooth is split again.
$Y = D_1 + D_2 + D_3 + S_3$

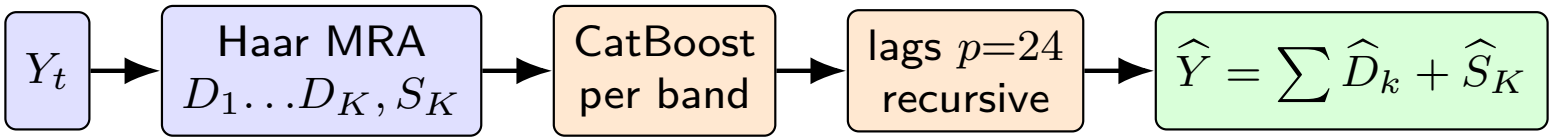
Inverse (IMODWT): same filters, reversed

$S_{j-1,t} = \sum_m p_m D_{j,t+2^{j-1}m} + \sum_m q_m S_{j,t+2^{j-1}m}$

Filter coefficients



WaveCatBoost pipeline



- Paper hybrid (Borah et al.): decompose $\rightarrow$ light tree per scale $\rightarrow$ sum.
- Same K and lag p for every Wave head; only the tree family changes.
- Point forecast alone is not enough for air-quality decisions — we need uncertainty (next slides).

Why CatBoost on each band?

Why a tree head

- Fast enough for near real-time AQ use.
- Handles nonlinear lag effects without a deep net.
- CatBoost: ordered boosting, strong default regularisation.
- Paper picks CatBoost over raw LGB / XGB / RNNs on MASE.

Why *one model per band*

- D_1: spiky, high-frequency noise.
- D_k: mid-scale diurnal swings.
- S_K: slow level / trend.
- Separate CatBoosts learn band-specific lag structure; a single raw model mixes all scales.

Family

WaveCatBoost = Haar MRA + CatBoost$(D_1), \ldots,$CatBoost(S_K).
WaveLGB / WaveXGB keep the same bands; only the fitter changes.

Features, recursive forecast, reconstruct

Features (each band). Lag vector of length p:

$$\mathbf{x}_t^{(k)} = \left(D_{k,t-1}, \ldots, D_{k,t-p}\right) \quad \text{(same for } S_K\text{)}.$$

No calendar / weather covariates in the letter WaveCatBoost head.

Recursive h-step

For $i = 1, \ldots, h$ on each band:

1. Build lag row from history.
2. Predict next value with CatBoost.
3. Append prediction; roll the window.

Multi-step uses *its own* forecasts as lags (honest bake-off protocol).

Reconstruct

Additive MRA $\Rightarrow$ just sum:

$$\widehat{Y}_{t+i} = \sum_{k=1}^{K} \widehat{D}_{k,t+i} + \widehat{S}_{K,t+i}.$$

Equivalent to IMODWT when bands are Haar MRA details + smooth.

Letter notebook also uses rolling **1-step** (observed lags) — that is Table 1, not the recursive bake-off.

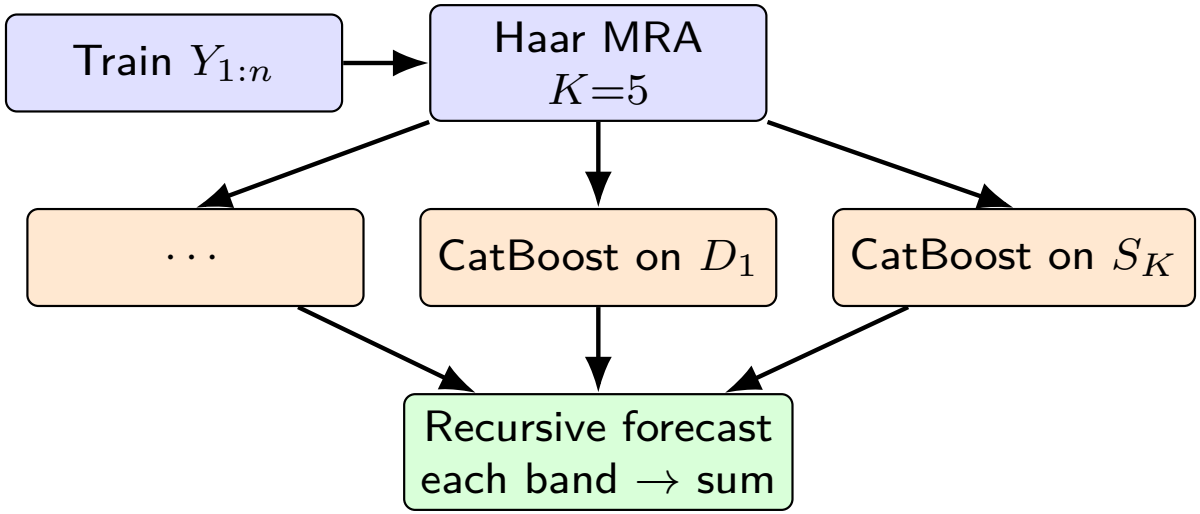
WaveCatBoost hyperparameters

Paper does not publish tree settings. We screen once on **CPCB PM$_{2.5}$** (train only, 1-day split) and freeze for every sensor / pollutant.

Setting	Frozen value
Wavelet filter	Haar
Depth K (J)	5 (capped; WaveLGB CV preferred 9)
Lag p	24
Tree depth	6
Learning rate η	0.05
Rounds / early stop	160 / 30

- Screening grid uses **LightGBM as a fast proxy**; winning $(p, \text{depth}, \eta, K)$ reused by WaveCatBoost.
- Letter-notebook 1-step run: CatBoost, lags 24, 1000 trees (separate from the recursive bake-off).

WaveCatBoost — training sketch



- Fit CatBoosts on **train bands only**; scaler also fit on train.
- Val (14 d) used only for the one-time hyperparameter screen.
- Test = last h hours; score MASE / MAE / RMSE after inverse scaling.

Error metrics (formulas)

With train length N, horizon h, seasonality $S{=}24$:

$$\mathrm{MASE} = \frac{\frac{1}{h}\sum_{t=1}^{h} |\widehat{Y}_t - Y_t|}{\frac{1}{N-S}\sum_{t=S+1}^{N} |Y_t - Y_{t-S}|}$$

(equivalent to paper Eq. (6) / Hyndman seasonal MASE).

$$\mathrm{MAE} = \frac{1}{h}\sum_{t=1}^{h} |\widehat{Y}_t - Y_t|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{h}\sum_{t=1}^{h} (\widehat{Y}_t - Y_t)^2}.$$

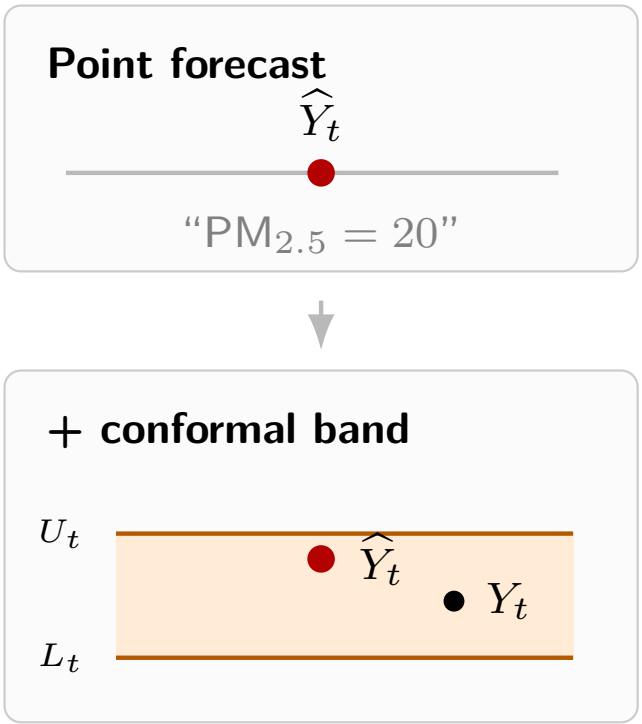
- MASE < 1: better than seasonal naive on the train scale.
- Lower is better for all three.

Our recursive bake-off (1-day MASE)

Sensor	Pol.	snaive	LGB	CatBoost	WaveCatBoost
CPCB	$PM_{2.5}$	2.02	0.74	0.82	**0.71**
CPCB	PM_{10}	1.46	0.52	0.75	0.53
CPCB	NO_2	0.72	0.52	0.51	0.55
ID1	$PM_{2.5}$	0.71	0.84	0.67	0.42
ID1	PM_{10}	0.66	0.81	0.73	0.41
ID1	NO_2	1.20	1.08	1.06	1.24

- **Protocol:** recursive multi-step from one origin (`scores.csv`).
- Fair for "does wavelet help vs raw CatBoost / LGB?" — **not** Table 1.

Why conformal prediction?



- **Need:** alerts need a safe range, not one risky guess.
- **Recipe:** use recent $|Y - \widehat{Y}|$ to set width — large recent errors $\Rightarrow$ wider band.
- **No normal assumption:** AQ errors are spiky / quiet; conformal is distribution-free.
- **"95%":** about 95 of 100 hours, $Y_t \in [L_t, U_t]$ (coverage).

From $\widehat{Y}_t$ alone to a band $[L_t, U_t]$ with target coverage $1 - \alpha = 95\%$.

Conformal formula

Score $CS_t = \dfrac{|Y_t - \widehat{Y}_t|}{\widehat{U}_t}$

Quantile $CQ_t = Q_{1-\alpha}\big(CS_{t-\kappa+1:t}\big)$

Interval $PI_t = \widehat{Y}_t \pm CQ_t\,\widehat{U}_t$

Band edges: $L_t = \widehat{Y}_t - CQ_t\widehat{U}_t$, $U_t = \widehat{Y}_t + CQ_t\widehat{U}_t$ (clip $L_t \geq 0$).

- Y_t, $\widehat{Y}_t$: truth / WaveCatBoost.
- $\widehat{U}_t$: local error scale (next slide).
- CS_t: scaled |error| (nonconformity).
- α=0.05 $\Rightarrow$ 95% target.
- $\kappa \in \{24, 168, 336\}$ h (cal. window).

How to use it

1. Fit on train; residuals on a cal. window.
2. Build CS_t; pick κ for $\approx 95\%$ cal. coverage.
3. Test: band $\widehat{Y}_t \pm CQ_t\widehat{U}_t$.
4. Coverage = share of $Y_t \in [L_t, U_t]$.

How $\widehat{U}_t$ and CQ_t are computed

$\widehat{U}_t$ — local error scale

Residual $e_t = Y_t - \widehat{Y}_t$.

Default: **24-h rolling MAE**

$$\widehat{U}_t = \text{mean}\left(|e_{t-23}|, \ldots, |e_t|\right).$$

Large recent mistakes $\Rightarrow$ large $\widehat{U}_t$ $\Rightarrow$ wider band via $CQ_t\widehat{U}_t$.

Multi-step test (no new Y yet): freeze one robust $u^\star$ from calibration and reuse it.

$CQ_t = Q_{1-\alpha}(CS_{t-\kappa+1:t})$

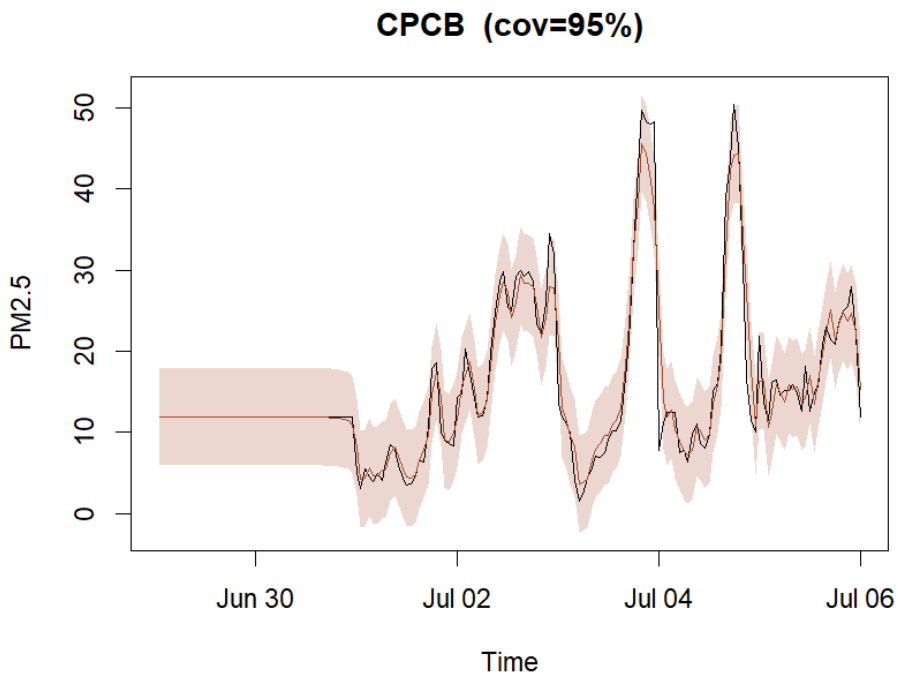
1. Take the last κ scores $CS_{t-\kappa+1}, \ldots, CS_t$.
2. Sort them.
3. Take the $(1-\alpha)$-quantile (α=0.05 $\Rightarrow$ $\approx$ 95th percentile).

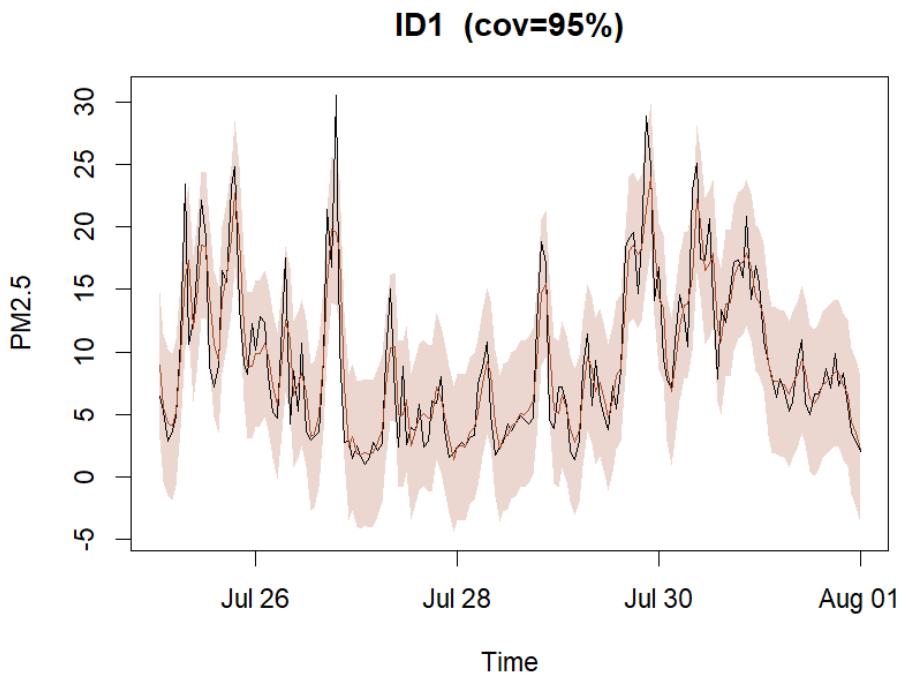
CQ_t is the cutoff for a "normal" scaled error.

Scores with $CS \leq CQ_t$ fall inside $\widehat{Y}_t \pm CQ_t\widehat{U}_t$.

Do the intervals work? (7-day test)

Purpose: letter-style 7-day $PM_{2.5}$ band — does truth sit inside the 95% window around WaveCatBoost?





- **Black** = actual; **orange** = WaveCatBoost; **shaded** = 95% residual PI $[L_t, U_t]$ (same colours as Rmd §8.2).
- Paper residual PI on this window: CPCB / ID1 coverage $\approx 95\%$ (widths ≈ 12.0 / 11.7).

7-day paper residual PI — summary

Sensor	Coverage	Mean width	Point MAE
CPCB	0.95	12.0	1.48
ID1	0.95	11.7	1.96
Shillong	0.95	19.3	3.56

Protocol

Rolling 1-step WaveCatBoost (letter notebook) + 95% quantile of $|Y - \widehat{Y}|$ on the plotted 7-day window ($\alpha = 0.05$).

Extensions — overview

Outside the letter bake-off. Filter / Fourier / EMD use a matched head and 1-day split (`analysis.Rmd` §10). Shillong A0–C3 is a separate CAAQMS architecture bake-off.

- **Other wavelet filters:** Haar vs D4 vs LA8 (WaveCatBoost).
- **Fourier bands** vs Haar MODWT (WaveLGB).
- **EMD** vs Haar MODWT (WaveLGB, $PM_{2.5}$).
- **Shillong A0–C3:** input / decomp designs on Lumpyngngad $PM_{2.5}$.

Haar vs D4 vs LA8 (WaveCatBoost, 1-day MASE)

Same WaveCatBoost head; only the MODWT filter (p_m, q_m) changes ($S_{0,t} = Y_t$, indices mod N):

$$D_{j,t} = \sum_{m=0}^{M-1} p_m \, S_{j-1,(t-2^{j-1}m) \bmod N}, \qquad S_{j,t} = \sum_{m=0}^{M-1} q_m \, S_{j-1,(t-2^{j-1}m) \bmod N}.$$

Haar (M=2) $q = (\frac{1}{2}, \frac{1}{2})$, $p = (\frac{1}{2}, -\frac{1}{2})$
D4 (M=4) $q \approx (0.483,\ 0.837,\ 0.224,\ -0.129)$, $p \approx (-0.129,\ -0.224,\ 0.837,\ -0.483)$
LA8 (M=8) $q \approx (-0.076,\ -0.030,\ 0.498,\ 0.804,\ 0.298,\ -0.099,\ -0.013,\ 0.032)$
$p \approx (0.032,\ 0.013,\ -0.099,\ -0.298,\ 0.804,\ -0.498,\ -0.030,\ 0.076)$

Sensor	Pol.	Haar	D4	LA8	Sensor	Pol.	Haar	D4	LA8
CPCB	NO_2	0.541	**0.510**	0.529	ID1	NO_2	**1.252**	1.346	1.330
CPCB	$PM_{2.5}$	**0.727**	0.909	0.955	ID1	$PM_{2.5}$	**0.421**	0.457	0.502
CPCB	O_3	**1.014**	1.212	1.119	ID1	O_3	**0.935**	0.940	0.964
CPCB	CO	**0.885**	0.993	1.200	ID1	CO	**0.798**	1.001	1.170
CPCB	SO_2	6.232	6.041	**5.972**	ID1	SO_2	**1.320**	1.390	1.379
CPCB	PM_{10}	0.535	**0.524**	0.538	ID1	PM_{10}	**0.368**	0.445	0.462

Haar best on **9/12** cells; D4 on 2; LA8 on 1 — Haar stays the default.

Fourier vs Haar MODWT (WaveLGB, 1-day)

Octave FFT bands matched to Haar layout (J=5, p=24). Global sinusoids — no local $D_{1,t}$ (`extension_scores`, Rmd §10.1).

Sensor	Pollutant	Haar	Fourier
CPCB	NO_2	**0.518**	0.692
CPCB	$PM_{2.5}$	**1.031**	1.272
ID1	NO_2	**1.259**	1.438
ID1	$PM_{2.5}$	**0.360**	0.473

- Haar wins every scored cell.
- Fourier smears spikes across frequencies; lags still carry a little of the event.

EMD vs Haar MODWT (WaveLGB, 1-day $PM_{2.5}$)

Adaptive IMFs folded into $D_1 \ldots D_5, S_5$ slots (Rmd §10.2). Same WaveLGB head as Haar / Fourier.

Sensor	Haar	EMD
CPCB $PM_{2.5}$	**1.031**	7.460
ID1 $PM_{2.5}$	**0.360**	0.616

Takeaway

Adaptive IMFs do *not* beat fixed Haar under a matched head. Unstable IMF counts / mode mixing hurt recursive band forecasts — keep EMD as a negative control.

Shillong A0–C3 — architecture bake-off

CAAQMS stations (mainly **Lumpyngngad**): different *input designs* for $PM_{2.5}$, not the letter WaveCatBoost head.

Code	Design
A0	Univariate — only $PM_{2.5}$'s own 24 lags
A1	A0 + weather (RH, wind, rain, . . .)
A2	Multivariate — 24 lags of all usable pollutants
A3	A2 + weather
B0–B3	Same ideas after Haar decomp (predict bands, then sum)
C1–C3	Decomposed + cross-band / cross-pollutant inputs

Shillong A0–C3 — Lumpyngngad takeaways

- **A2 beats A0:** other pollutants' lags help $PM_{2.5}$.
- **Decomposed B/C does not beat A2:** Haar on top of multivariate lags adds no gain at Lumpyngngad.
- Weather (A1 / A3) is secondary to the multivariate-lag signal.
- Contrast with the letter path: there Haar+CatBoost-per-band wins on CPCB/ID1; here multivariate raw lags win the Shillong bake-off.

Role in the deck

Extension only — does not replace the headline WaveCatBoost model.

Future work

Experiments we ran (extend / harden next):

- **Recursive bake-off:** WaveCatBoost vs snaive / LGB / CatBoost (1-day MASE, CPCB & ID1).
- **Letter residual PI:** 7-day 95% bands on WaveCatBoost (CPCB, ID1, Shillong).
- **Filter ablation:** Haar vs D4 vs LA8 under WaveCatBoost (12 sensor$\times$pollutant cells).
- **Fourier bands** vs Haar MODWT (WaveLGB, 1-day).
- **EMD** vs Haar MODWT (WaveLGB, $PM_{2.5}$).
- **Shillong A0–C3:** univariate / weather / multivariate / Haar-decomp / cross-band inputs on Lumpyngngad $PM_{2.5}$.
- **Next:** causal MODWT, more stations, tighter UQ, and push A2-style multivariate features into the letter WaveCatBoost path.

References I



J. Borah *et al.*, "WaveCatBoost for probabilistic forecasting of multivariate time series data," *IEEE Sensors Letters*, 2025.
Public code: WaveCatBoost-Time-Series-Model.



L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," *NeurIPS*, 2018.